

An independent set of size 1129 in the sixth strong power of C_7

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Abstract

We give an independent set of size 1129 in the sixth strong power of C_7 , improving the best known value 1120. A certificate and two independent verifiers are included. The best known lower bound on $\Theta(C_7)$ is unchanged.

1 Result

For a graph G the strong power $G^{\boxtimes n}$ has vertex set $V(G)^n$, and distinct vertices u, v are adjacent when for every i the coordinates u_i and v_i are equal or adjacent. For a cycle this means that distinct $u, v \in \mathbb{Z}_q^n$ are adjacent exactly when their cyclic distance is at most 1 in every coordinate. Independent sets of $C_q^{\boxtimes n}$ are therefore the codes in $\mathbb{Z}_q^n$ whose minimum cyclic Chebyshev distance is at least 2. The Shannon capacity is $\Theta(G) = \sup_n \alpha(G^{\boxtimes n})^{1/n}$.

Theorem 1. $\alpha(C_7^{\boxtimes 6}) \geq 1129$.

The certificate is the file `alpha-C7-6-1129.txt`, which lists 1129 vertices of $C_7^{\boxtimes 6}$, one per line, as six digits, the first digit being coordinate 0.

2 Context

Two values were recorded before this note. The first is $1101 = 367 \cdot 3$, the product of the size-367 set of Polak and Schrijver [5] with a maximum independent set of C_7 . The second is 1120, found by Itty, Rosin, Carstensen and Reichman [3] and listed in their Table 5. They note that it does not improve the capacity bound. Neither does 1129.

Since $\vartheta(C_7) = 7 \cos(\pi/7)/(1 + \cos(\pi/7)) = 3.31766720\dots$, the Lovász bound [4] gives $\alpha(C_7^{\boxtimes 6}) \leq \vartheta(C_7)^6 = 1333.51\dots$, and 1129 is 0.847 of that. The current record $\Theta(C_7) \geq 3.25883262\dots$ comes from a recursion in a high power [2]; a single sixth-power set would have to reach 1198 to match it, and $1129^{1/6} = 3.22687635\dots$

3 Verification

Two verifiers are included, `verify.mjs` for Node.js and `verify.py` for Python 3. Neither has dependencies and they share no code, so agreement between them is not evidence from a single implementation. Both use the naive $O(|S|^2 n)$ all-pairs test, which is slow but short enough to read; on 1129 vertices it takes under a second.

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```
node verify.mjs alpha-C7-6-1129.txt 7 6
python verify.py alpha-C7-6-1129.txt 7 6
```

Each prints OK: 1129 distinct vertices of C_7^6 , pairwise non-adjacent. Changing one coordinate of one vertex makes both report adjacent pairs and exit with a nonzero status.

4 The search

The set was found by local search over the 117 649 vertices of C_7^6 , starting from the 1120-vertex set of [3]. For each vertex outside the current set we track how many members it is adjacent to. A vertex adjacent to none is added. A vertex adjacent to exactly one member can be exchanged for that member; this keeps the size fixed but moves the set sideways and sometimes frees a further vertex. When no such move helps for a while, the set is perturbed rather than restarted. Several seeds reached 1128; one reached 1129. Both sets are included.

The starting set matters. Local search from a random start reached 953 to 969 over three seeds within the times we tried, against 1120 from the published start. A free search over abelian subgroups of $\mathbb{Z}_N^6$, quantised into $\mathbb{Z}_7^6$ as in [5], gave a conflict-free orbit of size 1092 at $N = 1092$, and taking unions of such orbits and running local search on them did not improve on 1092.

5 Local optimality

Fix a coordinate d and let $L_j = \{x \in S : x_d = j\}$. Vertices in layers j and j' can be adjacent only if $|j - j'| \leq 1$ in $\mathbb{Z}_7$, and if $|j - j'| = 1$ the d -coordinate is already close, so adjacency depends only on the other five coordinates. This gives two moves.

The first re-optimises a single layer. Layer j meets only L_{j-1} and L_{j+1} , so with those frozen the admissible rests are those avoiding $N[\text{rest}(y)]$ for $y \in L_{j-1} \cup L_{j+1}$, and the new layer is a maximum independent set of C_7^5 on that set. On the 1129-vertex set this gains nothing in any of the six directions.

The second re-optimises two adjacent layers together. What the argument of Baumert et al. [1] constrains is the pair sum: $L_j \cup L_{j+1}$ is independent in C_7^5 , so $2|S| \leq 7\alpha(C_7^5)$. Treating a pair jointly allows vertices to move between the two layers, which the single-layer move cannot do. Over 15 sweeps with 8 perturbations the size never went above 1129. Each perturbation forced six or seven vertices in, which evicts about forty since every vertex has $3^6 - 1 = 728$ neighbours, and the sweeps recovered exactly those forty.

The same pair move took a 105-vertex set in C_7^4 to 108 while this work was being done, so it is not a weak move. One difference should be recorded: in dimension four the layer subproblem lives in $\mathbb{Z}_7^3$ and was solved exactly, while in dimension six it lives in $\mathbb{Z}_7^5$, has around 11 000 vertices, and was solved heuristically from a warm start.

Data and code

The certificates, both verifiers and the search code are archived with this note and need only a stock Node.js 20 or Python 3.

References

- [1] L. D. Baumert, R. J. McEliece, E. Rodemich, H. C. Rumsey, R. Stanley and H. Taylor, *A combinatorial packing problem*, in: Computers in Algebra and Number Theory, SIAM–AMS Proc. IV (1971), 97–108.
- [2] *Strengthening recursive constructions for zero-error Shannon capacity*, arXiv:2608.30273.
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- [4] L. Lovász, *On the Shannon capacity of a graph*, IEEE Trans. Inform. Theory **25** (1979), 1–7.
- [5] S. C. Polak and A. Schrijver, *New lower bound on the Shannon capacity of C_7 from circular graphs*, Inform. Process. Lett. **143** (2019), 37–40; arXiv:1808.07438.